WEEKLY PROJECT REPORT

Project Title: AI-Based GST Compliance Helper for MSMEs using OCR and Automated Reconciliation
Role: Developer
Week: 2
Name: Bhoomika
SRN: R23EF048
Date: 27/03/2026

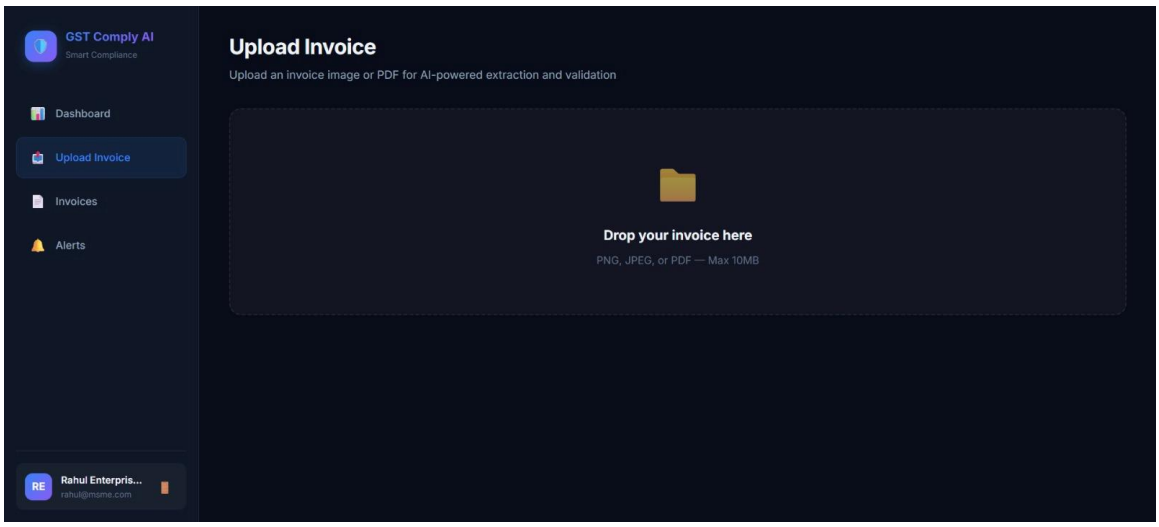
1. Objectives for the Week

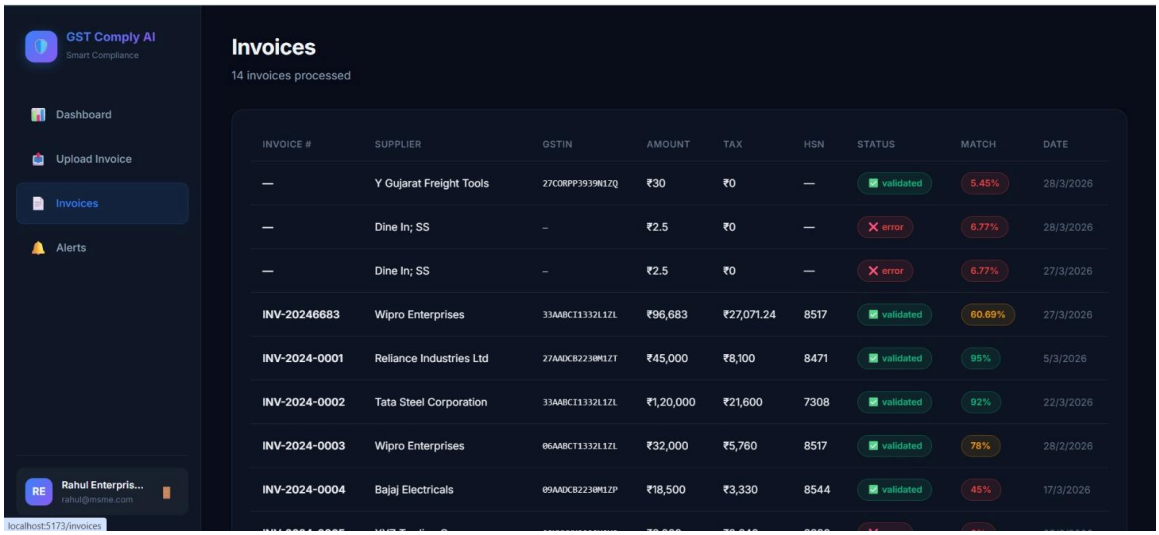
- To understand the finalized technology stack and assist in implementation
- To develop frontend modules and integrate them with backend APIs
- To work on OCR-based invoice upload and data extraction features

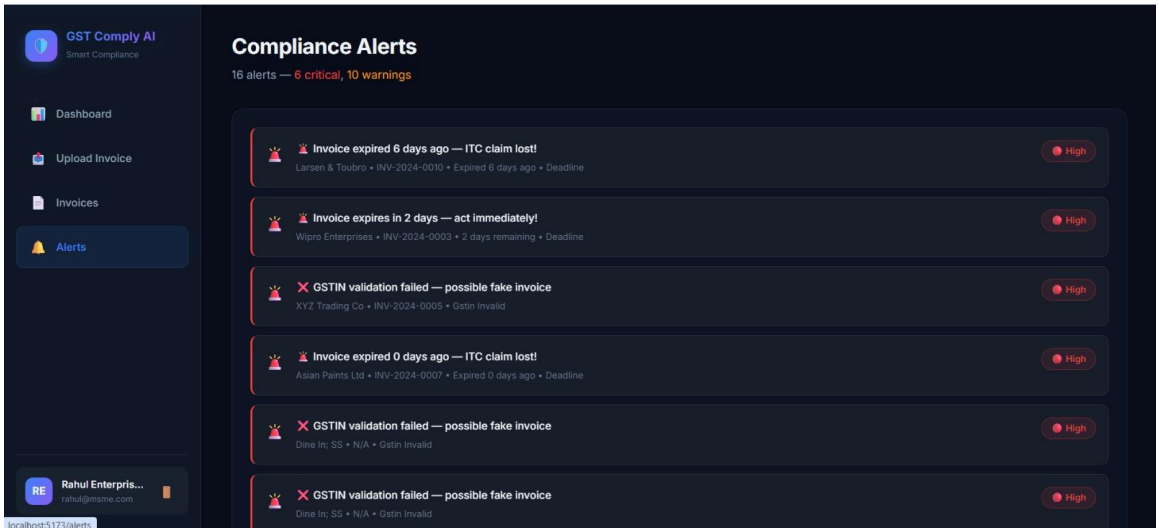
2. Activities Performed

- Worked with the finalized technology stack:
  Frontend: HTML, CSS, JavaScript
  Backend: Python (Flask/FastAPI)
  Database: SQLite
  OCR: pytesseract
  Image Processing: Pillow / OpenCV
  Data Extraction: Regex

- Developed frontend components (Dashboard, Upload, Results)
- Implemented invoice upload and API integration
- Displayed extracted invoice data
- Assisted in OCR testing and debugging


GST Comply AI
Smart Compliance
Dashboard
Upload Invoice
Invoices
Alerts
RE
Rahul Enterpris...
rahul@msme.com
Upload Invoice
Upload an invoice image or PDF for AI-powered extraction and validation
Drop your invoice here
PNG, JPEG, or PDF — Max 10MB


GST Comply AI
Smart Compliance
Dashboard
Upload Invoice
Invoices
Alerts
RE
Rahul Enterpris...
rahul@msme.com
localhost:5173/invoices
Invoices
14 invoices processed
INVOICE # SUPPLIER GSTIN AMOUNT TAX HSN STATUS MATCH DATE
— Y Gujarat Freight Tools 27CORPP3939N1ZQ ₹30 ₹0 — validated 5.45% 28/3/2026
— Dine In; SS – ₹2.5 ₹0 — error 6.77% 28/3/2026
— Dine In; SS – ₹2.5 ₹0 — error 6.77% 27/3/2026
INV-20246683 Wipro Enterprises 33AABCI1332L1ZL ₹96,683 ₹27,071.24 8517 validated 60.69% 27/3/2026
INV-2024-0001 Reliance Industries Ltd 27AADCB2230M1ZT ₹45,000 ₹8,100 8471 validated 95% 5/3/2026
INV-2024-0002 Tata Steel Corporation 33AABCI1332L1ZL ₹1,20,000 ₹21,600 7308 validated 92% 22/3/2026
INV-2024-0003 Wipro Enterprises 06AABCI1332L1ZL ₹32,000 ₹5,760 8517 validated 78% 28/2/2026
INV-2024-0004 Bajaj Electricals 09AADCB2230M1ZP ₹18,500 ₹3,330 8544 validated 45% 17/3/2026


GST Comply AI
Smart Compliance
Dashboard
Upload Invoice
Invoices
Alerts
RE
Rahul Enterpris...
rahul@msme.com
localhost:5173/alerts
Compliance Alerts
16 alerts — 6 critical, 10 warnings
Invoice expired 6 days ago — ITC claim lost!
Larsen & Toubro • INV-2024-0010 • Expired 6 days ago • Deadline
High
Invoice expires in 2 days — act immediately!
Wipro Enterprises • INV-2024-0003 • 2 days remaining • Deadline
High
GSTIN validation failed — possible fake invoice
XYZ Trading Co • INV-2024-0005 • Gstin Invalid
High
Invoice expired 0 days ago — ITC claim lost!
Asian Paints Ltd • INV-2024-0007 • Expired 0 days ago • Deadline
High
GSTIN validation failed — possible fake invoice
Dine In; SS • N/A • Gstin Invalid
High
GSTIN validation failed — possible fake invoice
Dine In; SS • N/A • Gstin Invalid
High

3. Key Design Decisions

- Simple UI for MSMEs
- Lightweight frontend
- Regex-based extraction
- Smooth frontend-backend communication

4. Contribution to Project Development

- Developed frontend modules
- Implemented upload and display
- Integrated OCR output
- Supported backend integration

5. Challenges Faced

- Handling OCR inconsistencies
- Extracting correct fields
- Managing missing data
- UI display issues

6. Learning Outcomes

- Learned frontend-backend integration
- Gained OCR experience
- Improved JavaScript and debugging skills

7. Plan for Next Week

- Improve OCR accuracy
- Enhance UI
- Integrate chatbot
- Improve reconciliation display

8. Conclusion

This week focused on frontend implementation and integration with backend. The system can now upload invoices, extract data, and display results.